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XIAMEN
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DIGITAL LOGIC

Chapter 3.1

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Outline:

- ※ **Logic Device Symbols**

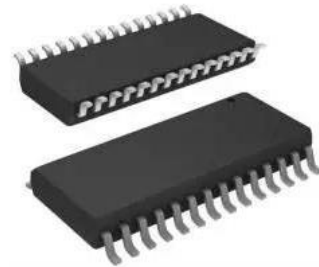
- ※ **Basic methods of combinational logic circuit analysis and design;**

- ※ **Several practical problems in the design of combinatorial logic circuits;**

- ※ **Problems of race and hazard in combinational logic circuits**

Logic devices such as integrated gates circuit and flip-flops are the material basis for implementing digital system functions.

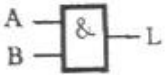
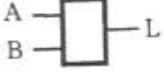
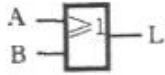
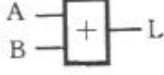
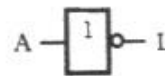
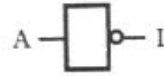
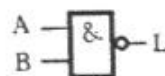
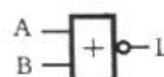
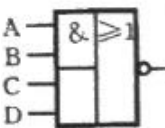
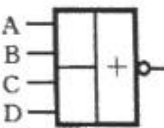
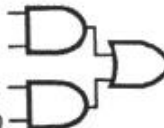
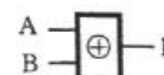
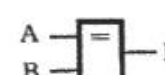
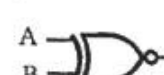
With the development of microelectronic technology, the components and wires of all kinds of logical functions are concentrated on the same semiconductor material. They are covered in a shell and connected by the pins to the outside world, that is, usually called **integrated circuit chips**.



The **advantages** of using integrated circuits for digital system design:

The advantages of high reliability, good dimensionality, low power consumption and low cost can greatly simplify the design and debugging process.

Common gate circuit symbols and expressions

Name	Logical function	New standard symbol	Still use symbol	Foreign popular symbol	Logical expression
AND gate	AND operator				$L = A \cdot B$
OR gate	OR operator				$L = A + B$
NOT gate	NOT operator				$L = \bar{A}$
NAND gate	NAND operator				$L = \overline{A \cdot B}$
NOR gate	NOR operator				$L = \overline{A + B}$
AND-OR-NOT gate	AND-OR-NOT operator				$L = \overline{AB + CD}$
XOR gate	XOR operator				$L = A \oplus B$ $= A\bar{B} + \bar{A}B$
XNOR gate	XNOR operator				$L = A \odot B$ $= \overline{A \oplus B}$ $= AB + \bar{A}\bar{B}$

II. Classification of integrated circuits according to their size

Generally, it is divided into SSI, MSI, LSI, and VLSI according to the number of logic gates or components included in a single integrated circuit chip.

1. SSI (Small Scale Integration) :

The number of logical gates is less than 10 gates (or the number of components is less than 100);

2. MSI (Medium Scale Integration) :

The number of logical gates is 10 to 99 gates (or the number of components is 100 to 999).

3. LSI (Large Scale Integration) :

The number of logical gates is 100 to 9999 gates (or the number of components is 1000 to 99999).

4. VLSI (Very Large Scale Integration) :

The number of logical gates is greater than 10000 gates (or the number of components is greater than 100000).

Logic gate circuit



Logical devices that implement basic logic operations and commonly used complex logic operations are collectively referred to as logic gate circuits, which are the basic unit circuits that make up the digital system.

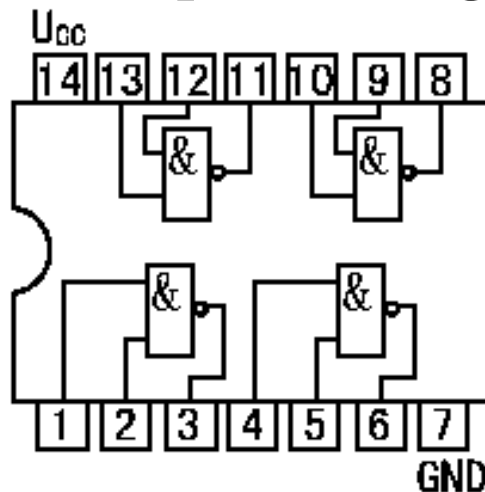
Learning should focus on the function and external features of integrated logic gates, as well as the usage of the device. It is only required to have a general understanding of its internal structure and working principles.

TTL NAND gate Integrated Circuit (IC) chip

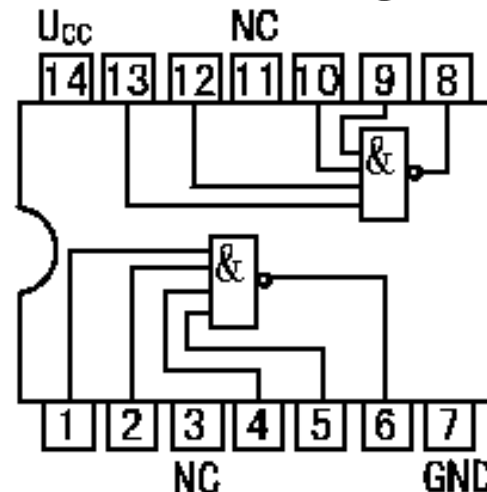
There are many types of TTL NAND gate IC chips, and commonly used TTL NAND gate IC chips have 7400 and 7420.

The 7400 pinout diagram is shown in Figure (a) .

The 7420 pinout diagram is shown in Figure (b) .



(a)



(b)

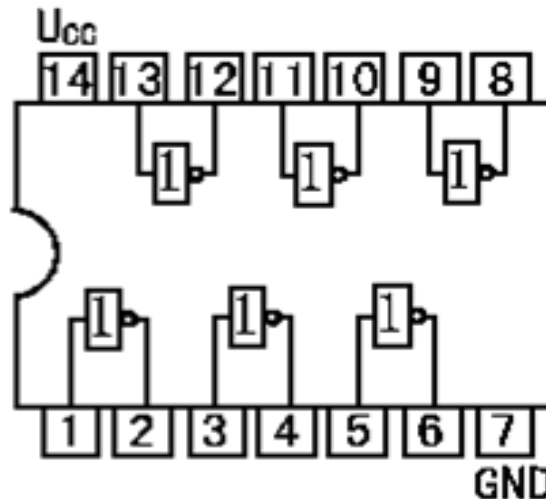
TTL7400、7420 pinout diagram

II. Other functions of the TTL gate circuit

The integrated TTL gate also has different functions such as **AND**, **OR**, **NOT**, **NOR**, **NOR**, and **XOR**. In addition, there are two special types of gate circuits—**open collector gates (OC gates)** and **tri-state gates (TS gates)**.

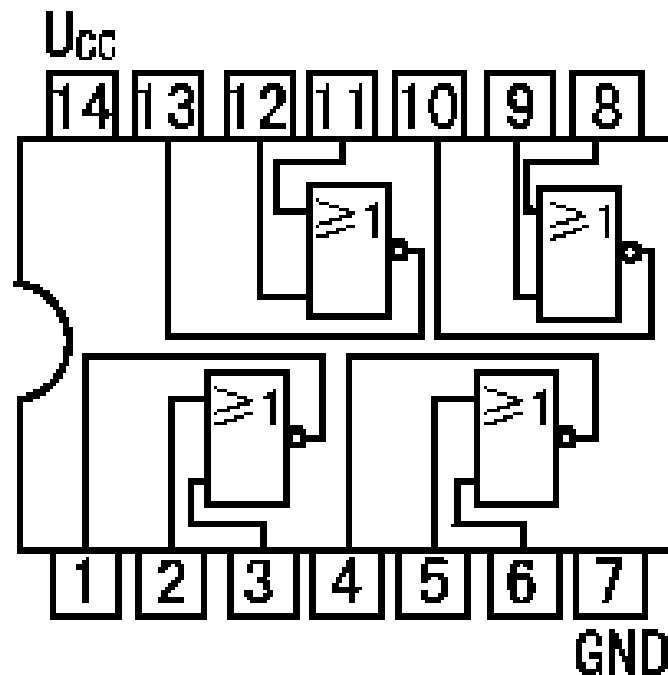
1. Several common TTL gate circuits

(1) Not Gate



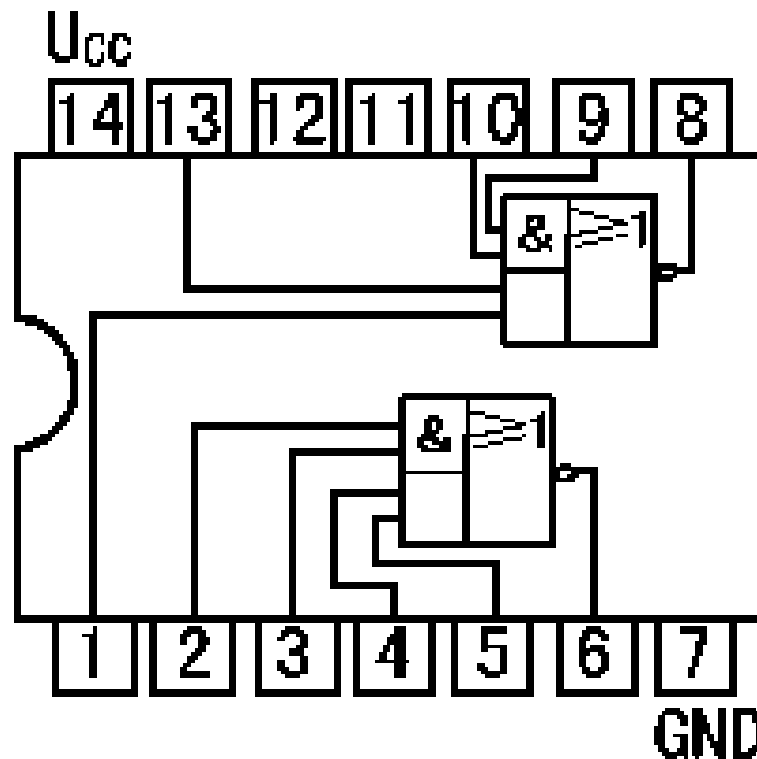
(2) NOR gate

Commonly used TTL NOR gate integrated circuit chip has 2 input 4 NOR gate 7402 and so on. The 7402 pin distribution diagram is shown in the following figure.



(3) AND-OR-INVERT gate

The pin arrangement diagram of commonly used TTL AND-OR-NOT gate IC chip 7451 is shown in the figure below.



3.2 Basic concepts of Combinational logic circuits

I. Definition

If the logic circuit produces a stable output value at any time depends only on the combination of input values at that time, and regardless of the past input value, it is called a combinational logic circuit.

II, Structure



III, Representation

A set of logical function expressions can be used to describe its logical functions. The function expression can be expressed as:

$$F_i = f_i (X_1, X_2, \dots, X_n) \quad i = 1, 2, \dots, m$$

IV. Features

The combinational circuits contains two features:

- ① A combinational circuit consists of a number of logic gate circuits, but does not contain any memory devices;
- ② The signal is transmitted in one direction and there is no feedback loop.

4. 2 Analysis of combinational logic circuits

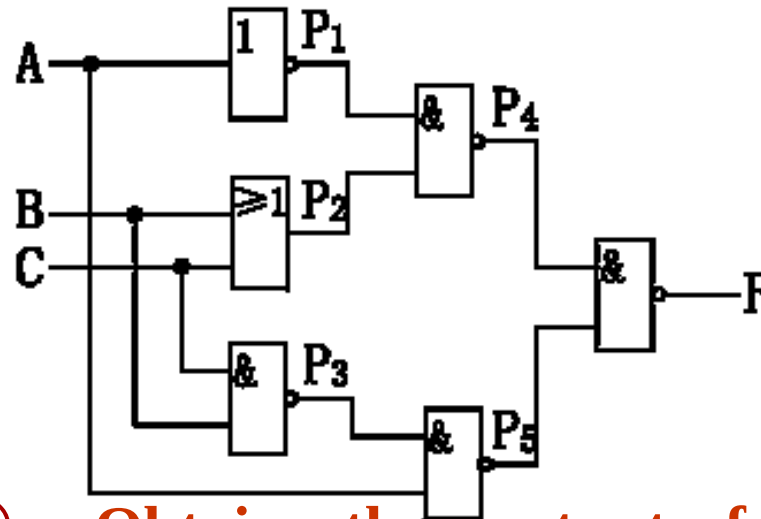
The logic circuit analysis refers to the logical relationship between a given logic circuit's input and its output.

I. Steps

- 1. Obtain the output function expression according to the logic circuit diagram**
- 2. Simplify output function expressions**
- 3. List the output function's truth table**
- 4. Evaluate the function**

II, Example

Analyze the combinational logic circuit shown in the figure below.



Solution: ① Obtain the output function expression according to the logic circuit diagram

$$\overline{A} = P_1 \quad P_2 = B + C \quad P_3 = \overline{B} \cdot \overline{C}$$

$$P_4 = \overline{P_1} \cdot P_2 = \overline{\overline{A}}(B + C) \quad P_5 = \overline{A} \cdot P_3 = \overline{A} \cdot \overline{\overline{B} \cdot \overline{C}}$$

$$F = \overline{P_4} \cdot P_5 = \overline{\overline{\overline{A}}(B + C)} \cdot \overline{\overline{A} \cdot \overline{\overline{B} \cdot \overline{C}}}$$

② **Simplify output function expressions:** using the logic algebra method to simplify F

$$\begin{aligned} F &= \overline{P_4} \cdot \overline{P_5} = \overline{\overline{A(B+C)} \cdot \overline{ABC}} \\ &= \overline{A(B+C)} + \overline{ABC} \\ &= \overline{A}B + \overline{A}C + A\overline{B} + A\overline{C} \\ &= A \oplus B + A \oplus C \end{aligned}$$

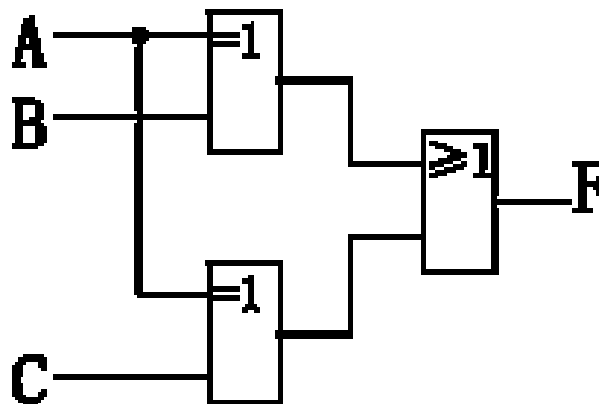
③ **List the truth table**
The truth table of function is:

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

④ Evaluate the design

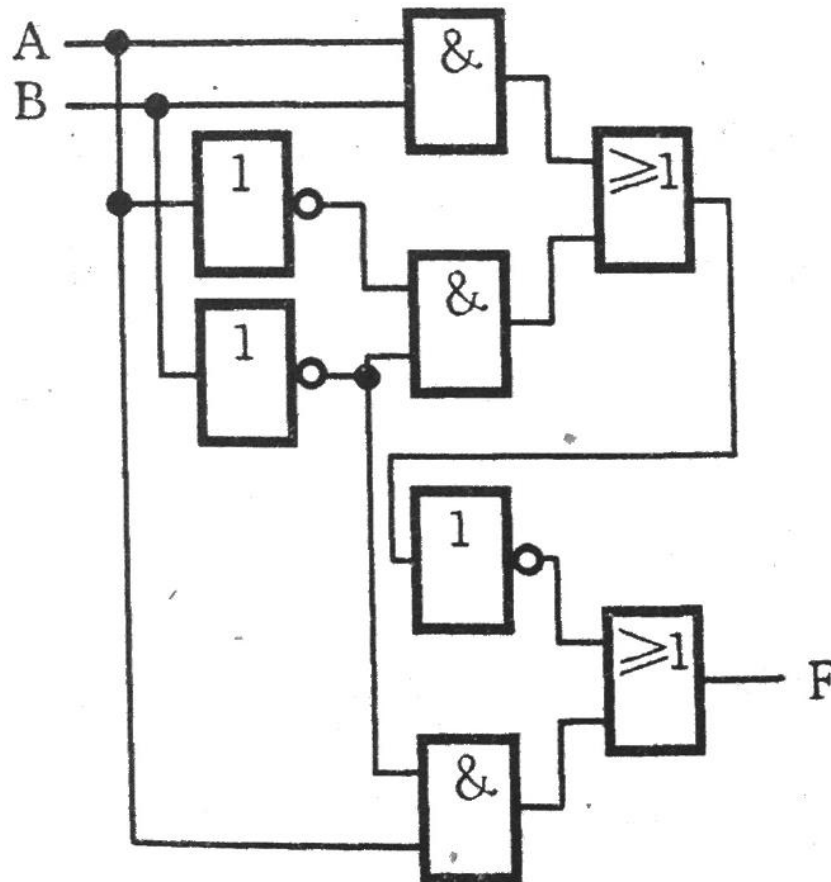
From the truth table, the circuit has a logic function to check if the input signal values are consistent. Once the output is 1, it indicates that the input is inconsistent. This circuit is often referred to as an "inconsistent circuit."

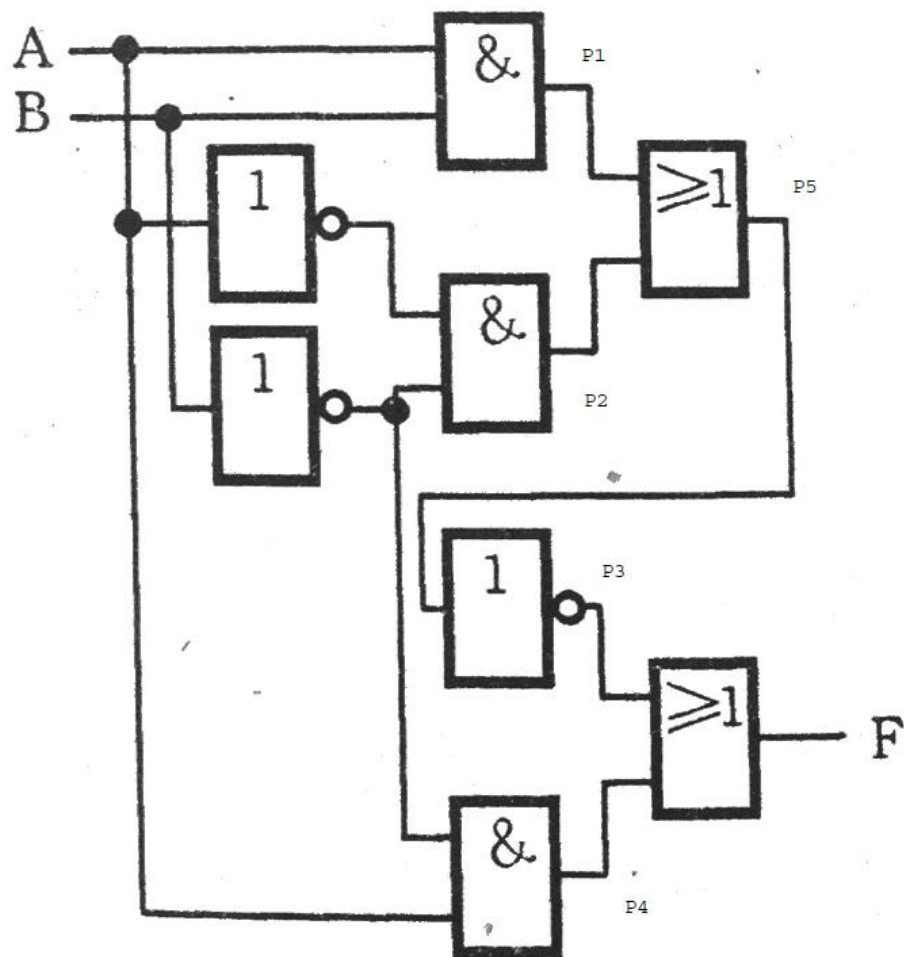
From the analysis, we can see that the circuit design scheme is not the simplest. According to the simplified output function expression, an XOR gate and an OR gate can be used to draw the logic diagram of the given function.



Practice

- Analyze the combinational logic circuit shown in the figure below.





$$P1 = AB$$

$$P2 = \bar{A} \bar{B}$$

$$P4 = A\bar{B}$$

$$P5 = P1 + P2 = AB + \bar{A} \bar{B}$$

$$P3 = \overline{P5} = \overline{AB + \bar{A} \bar{B}}$$

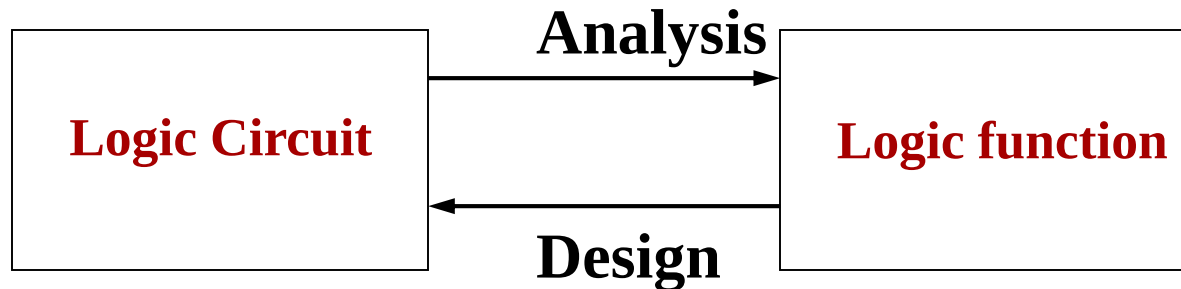
$$F = P3 + P4 = \overline{AB + \bar{A} \bar{B}} + A\bar{B}$$


$$\lceil F = A\bar{B} + \bar{A}B$$

- What is the logic function of this diagram?
- How about the design quality?

3.3 Combinational logic circuit design

According to the logic function that users ask to implement, designing process of the logic circuit that realizes the given function under the particular conditions, is called logic design, also called logic synthesis.



4.3.1 Design Method Overview

Since the various design requirements proposed in practical applications are generally described in text form, the primary task of logic design is to abstract the design requirements for text descriptions into a logical relationship. For a combinational logic circuit, the logical expression describing the problem is abstracted.



1. Establish a logical description of a given problem

The key to this step is to figure out the input and output of the circuit, establish the logical relationship between the input and output, and get a logical expression that describes the given problem.

There are two common methods for finding logical expressions, namely the truth table method and analysis method.

2. Find the simplest expression of the logic function

In order to minimize the number of logic gates included in the logic circuit and to minimize the number of wires, simplify the logic expression and find the simplest expression that describes the design problem.

- 3. Select the logic gate type and transform the logic function into the corresponding form
 - According to the simplified logical expression and the specific requirements of the problem, an appropriate logic gate is selected, and the logical expression is transformed into a form corresponding to the selected logic gate.
 - 4. Draw a logic circuit diagram
-
- Depending on how difficult it is and how proficient the designer is, sometimes some of these steps can be skipped. The design process can be flexibly controlled according to the specific situation.

4.3.2 Design Example

Question: Design a three-variable "majority voting circuit."

Solution: **Analysis:** The "majority voting circuit" is to vote on a certain resolution according to the principle of the minority obeying the majority, and to determine if it is passed.

Let the logical variables A , B , C --- represent the three members participating in the voting, respectively, and define that 0 stands for reject and 1 stands for recommend;

Logical function F ---- indicates the result of the voting. A value of 0 for F means that the resolution was denied, and a value of 1 for F means that the resolution was passed.

According to the principle of the minority obeying the majority, the relationship between the function and the variable is: When two or more of the three variables A , B , and C take a value of 1, the value of the function F is 1, and in other cases the function F value is 0.

① Establish a logical description of a given problem

Assuming the "Truth Table Method" is used, the truth table can be made as shown in the following table.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Use the truth table, we can obtain the
SOM expression of F:

$$F(A,B,C) = \sum m (3,5,6,7)$$

② 2. Find the simplest expression of the logic function

Draw the K map of $F(A,B,C) = \sum m(3,5,6,7)$ as

AB \ C	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Use the K map to obtain the simplest “AND-OR” expression is

$$F(A, B, C) = AB + AC + BC$$

③ Select the logic gate type and transform the logic function into the corresponding form.

Assuming that the NAND gate chip is used to implement a circuit for a given function, the above expression should be transformed into a “NAND-NAND” expression. which is

$$F(A, B, C) = \overline{\overline{AB + AC + BC}} = \overline{\overline{AB} \cdot \overline{AC} \cdot \overline{BC}}$$

④ 4. Draw a logic circuit diagram

From the function's "NAND-NAND" expression, the logic circuit diagram to achieve a given function can be drawn as shown in the figure below.

